

Case Study Part 1: Menopausal Hormone Therapy

Overview

- ▶ What is menopause?
- ▶ What is Menopausal Hormone Therapy?
- ▶ Status of research prior to Women's Health Initiative
- ▶ The Women's Health Initiative
- ▶ Risk as an estimand
- ▶ Recent observational studies

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- ▶ What happens?
 - ▶ How much did you know about menopause before reading?

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- ▶ diminished sleep quality

- ▶ insomnia

- ▶ irritability

- ▶ depression

- ▶ mood swings

- ▶ difficulty concentrating

- ▶ reduced quality of life

- ▶ aching joints

- ▶ bloating

- ▶ breast tenderness

- ▶ bladder/urinary discomfort

- ▶ more frequent urinary tract infections

- ▶ vaginal dryness, discharge, and bleeding

- ▶ loss of sexual desire, painful intercourse

- ▶ headaches

- ▶ thinning hair

- ▶ heart palpitations

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- ▶ 1980s: Progesterone added for women with uterus as estrogen alone increased rates of uterine cancer and endometrial cancer (estrogen alone still used for those without uterus)

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- ▶ 1980s-1990s: multiple observational studies suggest hormones (estrogen or estrogen+progesterone) reduce risk of heart disease, reduce development of osteoporosis, might reduce the risk of Alzheimer's, and are the best available treatment for vasomotor symptoms
 - ▶ “...estrogen should be in the water... we thought it was like fluoride” -Hadine Joffe, Harvard Medical School professor of psychiatry

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- ▶ Study halted July 2002 (average follow-up 5.2 years)

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- ▶ What do we mean by increased risks?
 - ▶ In 1 year for 10,000 postmenopausal women taking estrogen+progesterone, 8 more will have invasive breast cancer, 7 more will have a heart attack, 8 more will have a stroke, 18 more will have blood clots, than a similar group of 10,000 women not taking the hormones (Dr. Rossouw, acting director WHI)

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 - ▶ contextualization of risks

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 - ▶ Hazard ratio of treatment D :
$$\frac{\lambda(t|D=1)}{\lambda(t|D=0)} = \frac{\lambda_0(t) \exp(\beta_1 \cdot 1)}{\lambda_0(t) \exp(\beta_1 \cdot 0)} = \exp(\beta_1)$$

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- ▶ In 2002, the WHI Data and Safety Monitoring Board calculated an HR of 1.26, a 26% increase in breast cancer risk under treatment
- ▶ The WHI and media reporting on the halt of the study have been criticized for failing to communicate the difference between absolute and relative risks ($\frac{8}{10000} \neq .26$)

Results from WHI (and WHIMS)

What is the effect of estrogen+progesterone on invasive breast cancer in post-menopausal women?

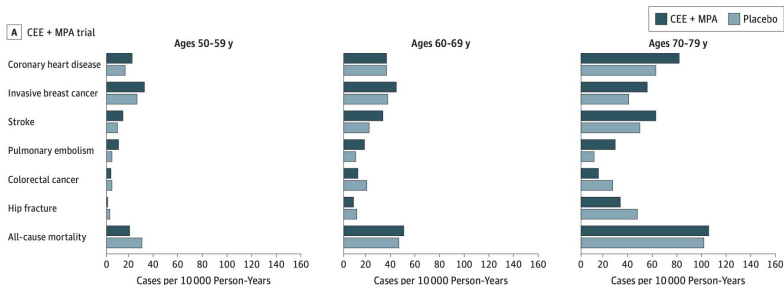
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- ▶ Intervention phase (1993-2002): HR of 1.24 (95% CI, 1.01-1.53)
- ▶ 9 more cases of invasive breast cancer per 10,000 person-years

Figure 3. Absolute Risks of Health Outcomes by 10-Year Age Groups in the Women's Health Initiative Hormone Therapy Trials During the Intervention Phase



Results from WHI (and WHIMS)

- Cumulative follow-up: HR 1.28 (95% CI, 1.11-1.48); risk attenuation with time since cessation of hormone therapy use

CEE + MPA Trial					
		No. (%) of Events ^a			
		CEE + MPA (n= 8506)	Placebo (n= 8102)	Difference/ 10 000 PY ^b	HR (95% CI)
Primary end points					
Coronary heart disease					
50-59 y	93 (0.26)	69 (0.21)	5	1.27 (0.93-1.74)	
60-69 y	201 (0.44)	199 (0.46)	-2	0.97 (0.79-1.18)	
70-79 y	193 (0.98)	162 (0.84)	14	1.17 (0.95-1.44)	
Invasive breast cancer					
50-59 y	132 (0.37)	93 (0.28)	9	1.34 (1.03-1.75)	
60-69 y	198 (0.43)	149 (0.34)	9	1.27 (1.02-1.57)	
70-79 y	104 (0.53)	81 (0.42)	11	1.25 (0.94-1.67)	

Cumulative Follow-Up

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- ▶ We still have longitudinal challenge: first-order autoregressive structure

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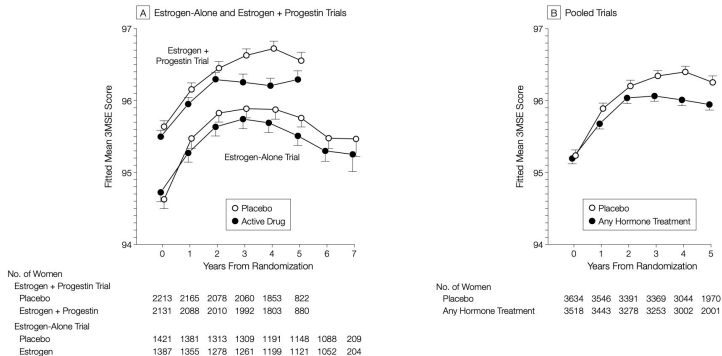
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- ▶ Difference is "too small to have relevance in clinical practice"

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Figure 2. Fitted Mean Modified Mini-Mental State Examination Scores for Estrogen-Alone and Estrogen Plus Progestin Trials and Pooled Trials



Recent Observational Studies

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 - ▶ Different women take different formulations/doses of hormones

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Table 2. Hormone Use Among Women With Breast Cancer and Women in the Control Group

Type of Hormone	Control Group (n=431,830)	Case Group (n=43,183)	Crude OR (95% CI)	Adjusted OR (95% CI)
Ever any menopausal HT*	62,753 (14.5)	7,280 (16.9)	1.19 (1.16–1.23)	1.12 (1.09–1.15)
Estrogen [†]				
No estrogen	373,019 (86.4)	36,391 (84.3)	1.0 (Ref)	1.0 (Ref)
Bioidentical	21,579 (5.0)	2,487 (5.8)	1.18 (1.13–1.23)	1.04 (1.00–1.09)
Animal-derived	28,184 (6.5)	3,311 (7.7)	1.20 (1.16–1.25)	1.01 (0.96–1.06)
Both	9,048 (2.1)	994 (2.3)	1.13 (1.05–1.20)	0.96 (0.89–1.03)
Progestogen [‡]				
No progestogen	410,256 (95.0)	40,352 (93.4)	1.0 (Ref)	1.0 (Ref)
Bioidentical	125 (0.03)	12 (0.03)	0.98 (0.54–1.77)	0.99 (0.55–1.79)
Synthetic	21,434 (5.0)	2,817 (6.5)	1.34 (1.28–1.39)	1.28 (1.22–1.35)
Both	15 (0.0)	2 (0.0)	—	—

Which of these is comparable to WHI?

Recent Observational Studies

- ▶ Kim et al. 2021: Association between menopausal hormone therapy and risk of neurodegenerative diseases: Implications for precision hormone therapy
 - ▶ Data on 1.4 million women from 2007 to 2016 on medical insurance (Humana)
 - ▶ 379,352 women met age (over 45) and (propensity score) matching criteria

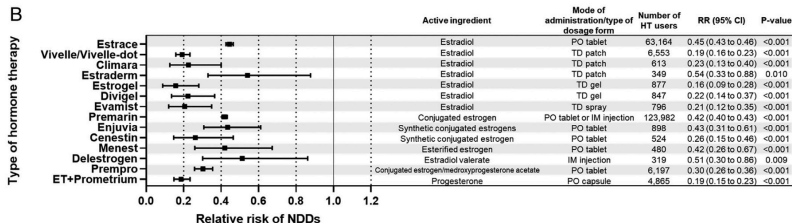
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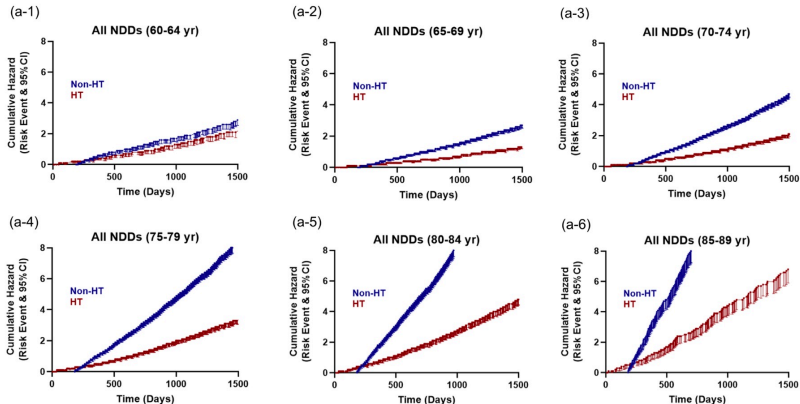
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Kim et al. 2021 also analyzes duration of use and administration (oral versus transdermal patch)

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- ▶ Analyze data based on WHI and WHIMS for the effect of CEE+MPA on breast cancer and cognition score
- ▶ Later: read about the effect of WHI/WHIMS on perceptions and prescriptions
- ▶ Later: analyze WHI- and WHIMS-based data with covariates
- ▶ Later: look closer at these recent observational studies